The direction of such eddy currents is so as to oppose the cause producing it which is the relative motion between disc and field. Thus it produces an opposing torque so as to reduce the oscillations of pointer. This is the most efficient method of damping.

6.4 Types of electrical measuring instruments based on principle of operation

1. Moving coil type
   (a) Permanent magnet moving coil instruments
   (b) Dynamometer type instruments
2. Moving iron type
   (a) Attraction type
   (b) Repulsion type

6.4.1 Permanent Magnet Moving Coil (PMMC) instruments

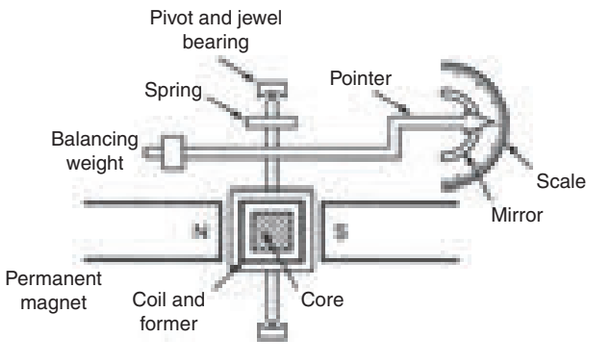


Fig. 6.6 Permanent magnet moving coil instrument

The permanent magnet moving coil instrument is shown in the figure 6.6. It is the most accurate type of instrument and is used for DC measurements.

The moving coil is wound with many turns of fine enameled or silk covered copper wire. The coil is mounted on rectangular aluminium former which is pivoted on jeweled bearings. The coil is placed in the magnetic field of a permanent magnet in the shape of a horse-shoe. Two spiral shaped phosphor bronze hair springs are fixed. They are used to carry the coil current and give the controlling torque to the coil. The damping torque is provided by eddy current damping. It is obtained by movement of the aluminium former, moving in the magnetic field of the permanent magnet.

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It works on the principle that when a current carrying coil (or conductor) is placed in the magnetic field, the coil (or conductor) experiences a force and moves. The amount of force experienced by the coil is proportional to the magnitude of current through the coil.

Thus deflecting torque is directly proportional to the current passing through the coil and hence the scale is uniform. The direction of deflection is according to the direction of current through the coil.

(A) Advantages

i. The scale is uniform
ii. No hysteresis loss
iii. Low power consumption
iv. No effect of stray magnetic field
v. High torque-weight ratio
vi. Effective and efficient eddy current damping
vii. Range can be extended by using shunts and multipliers

(B) Disadvantages

i. Can be used for DC measurements only.
ii. Cost is higher than that of moving iron instruments.

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